VARMA simulation results

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Complete run: 12,000 saved replications; 120,000 attempted estimates; 4 designs.

Sample sizes: 100, 200, 400. Planned replications per cell: 1000. Seed: 20261110.

Recorded compute/checkpoint elapsed time: 1.41 hours; worker counts: 7.

MLE backend: skf; initial covariance: stationary; pilot: HR1. CML and the echelon estimators use conditional Gaussian likelihood.

Primary results retain every finite loss, including extreme outputs and nonconverged estimates.

Qualified results require stable AR and invertible MA, irrespective of optimizer convergence.

NET's convergence flag concerns its IHR pilot; HR1, DPF and VAR (BIC) have no iterative flag.

The paper-style trimmed comparison removes 1.0% of replication losses from each tail.

Neither availability nor qualification makes these unconditional risks of an algorithm that can fail.

IRF bands are Monte Carlo sampling quantiles, not confidence intervals.

MCSEs use independent replication-level losses. Heavy tails can make them unreliable.

Relative errors pair each method with its benchmark on common usable draws.

This run evaluates IRFs and point forecasts. It does not evaluate FEVDs, bootstrap coverage,
or prediction-interval coverage. These require additional experiments.

Diagnostics and index selection

CML: 0 forecast failures; 1 nonconverged; 12000 qualified / 12000 attempted.

DGP: 0 forecast failures; 0 nonconverged; 12000 qualified / 12000 attempted.

DPF: 0 forecast failures; 0 nonconverged; 11977 qualified / 12000 attempted.

ECH oracle: 0 forecast failures; 0 nonconverged; 12000 qualified / 12000 attempted.

ECH selected: 0 forecast failures; 0 nonconverged; 12000 qualified / 12000 attempted.

HR1: 0 forecast failures; 0 nonconverged; 11973 qualified / 12000 attempted.

IHR: 0 forecast failures; 6650 nonconverged; 10586 qualified / 12000 attempted.

MLE: 0 forecast failures; 1 nonconverged; 12000 qualified / 12000 attempted.

NET: 0 forecast failures; 6650 nonconverged; 11452 qualified / 12000 attempted.

RBY: 0 forecast failures; 338 nonconverged; 12000 qualified / 12000 attempted.

VAR (BIC): 0 forecast failures; 0 nonconverged; 11999 qualified / 12000 attempted.

Design 1: 90.4% exact index recovery at n = 400 (true vector: 1,0).

Design 2: 93.6% exact index recovery at n = 400 (true vector: 1,1).

Design 3: 44.1% exact index recovery at n = 400 (true vector: 1,1).

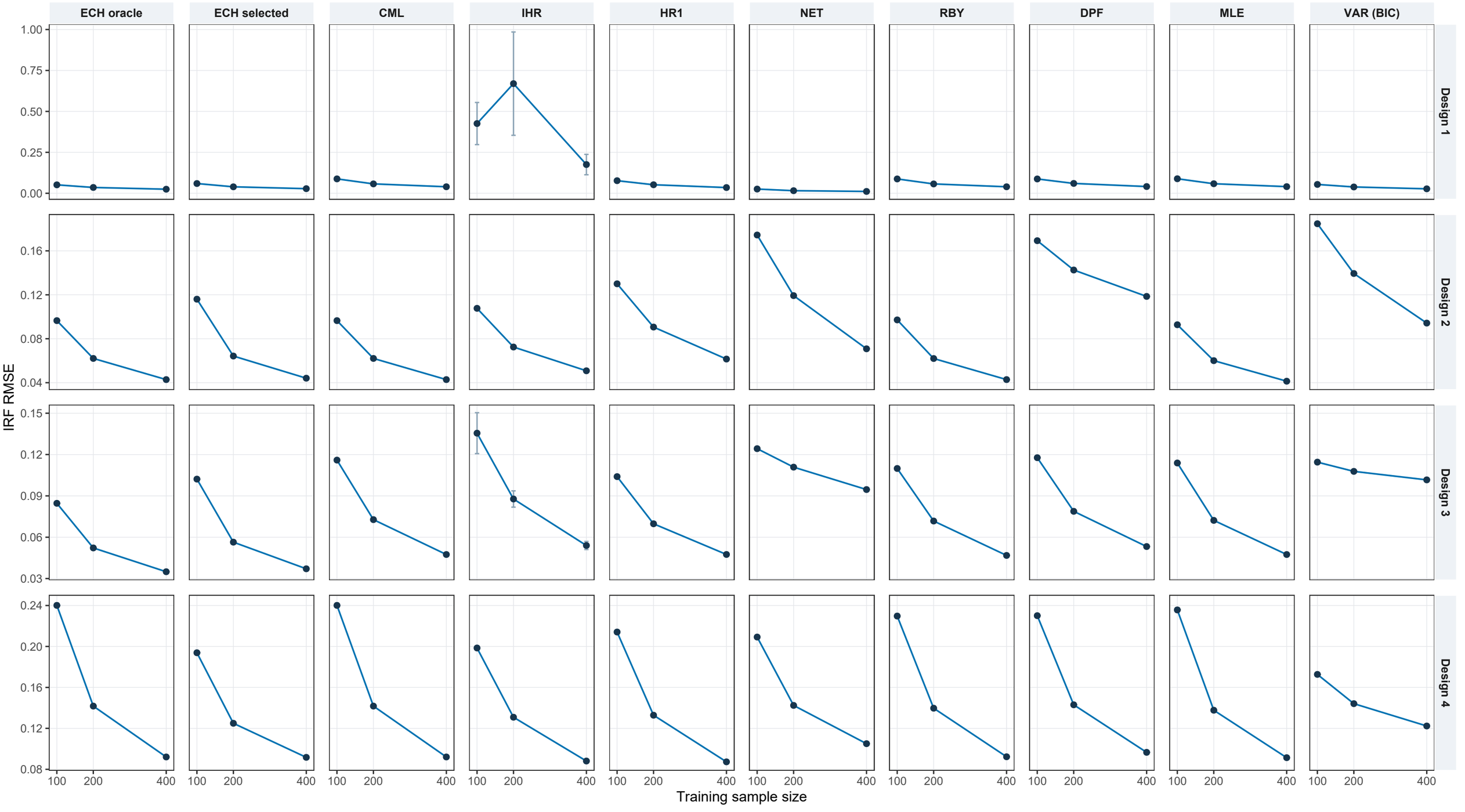
Design 4: 0.8% exact index recovery at n = 400 (true vector: 2,2,2).

Counts refer to all saved replications; categories overlap. NET uses its IHR pilot's convergence flag.

The figures and tables include both all available outputs and the separately qualified subset.

Transfer-function accuracy

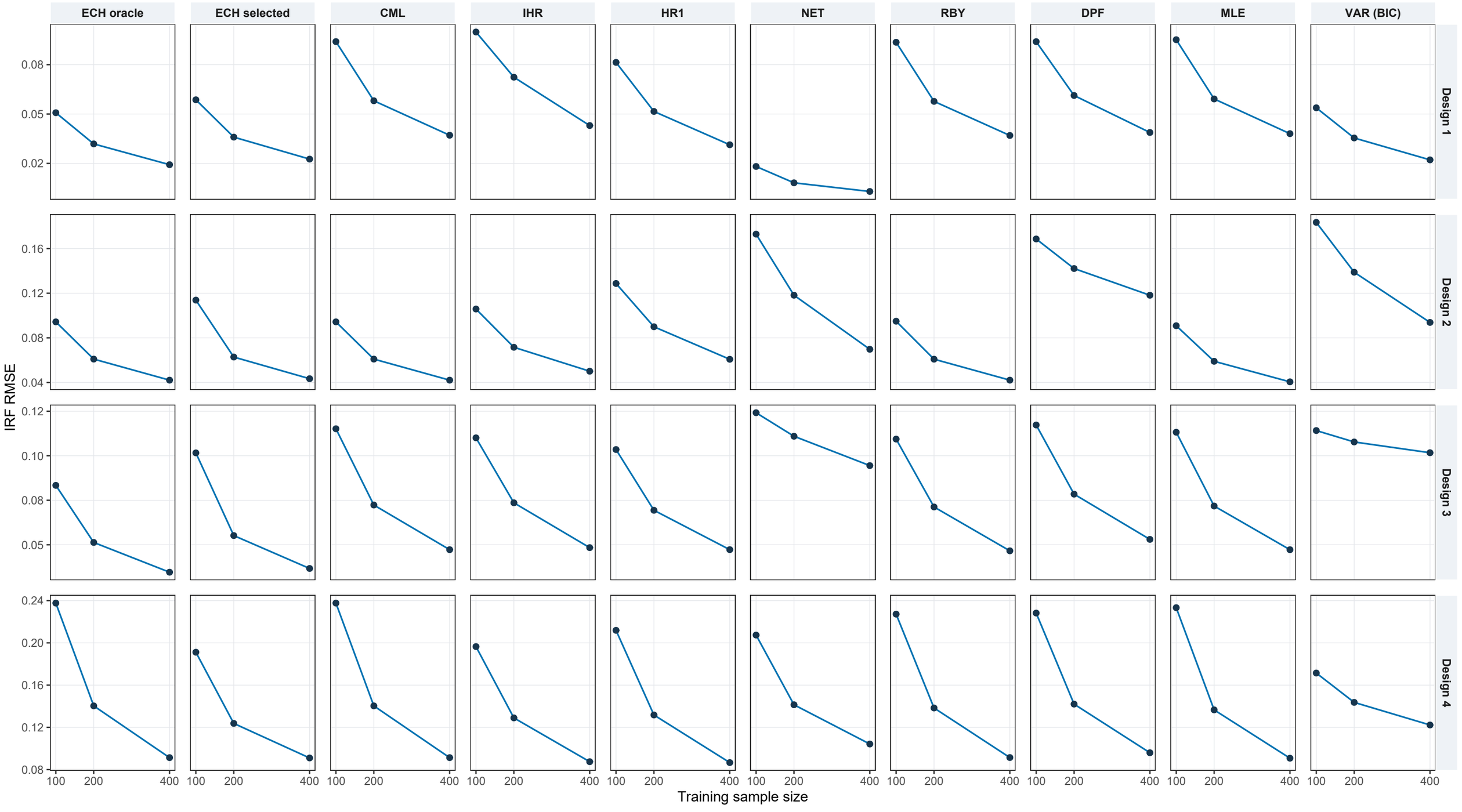
All available estimates | reduced-form IRFs, lags 1-5 | 1000 planned replications per cell



Untrimmed RMSE; whiskers show one Monte Carlo standard error (lower ends floored at zero).
Entries and lags are pooled within each replication. Lines do not impose a convergence rate. MCSEs can be unstable with extreme losses.

Transfer-function accuracy: trimmed comparison

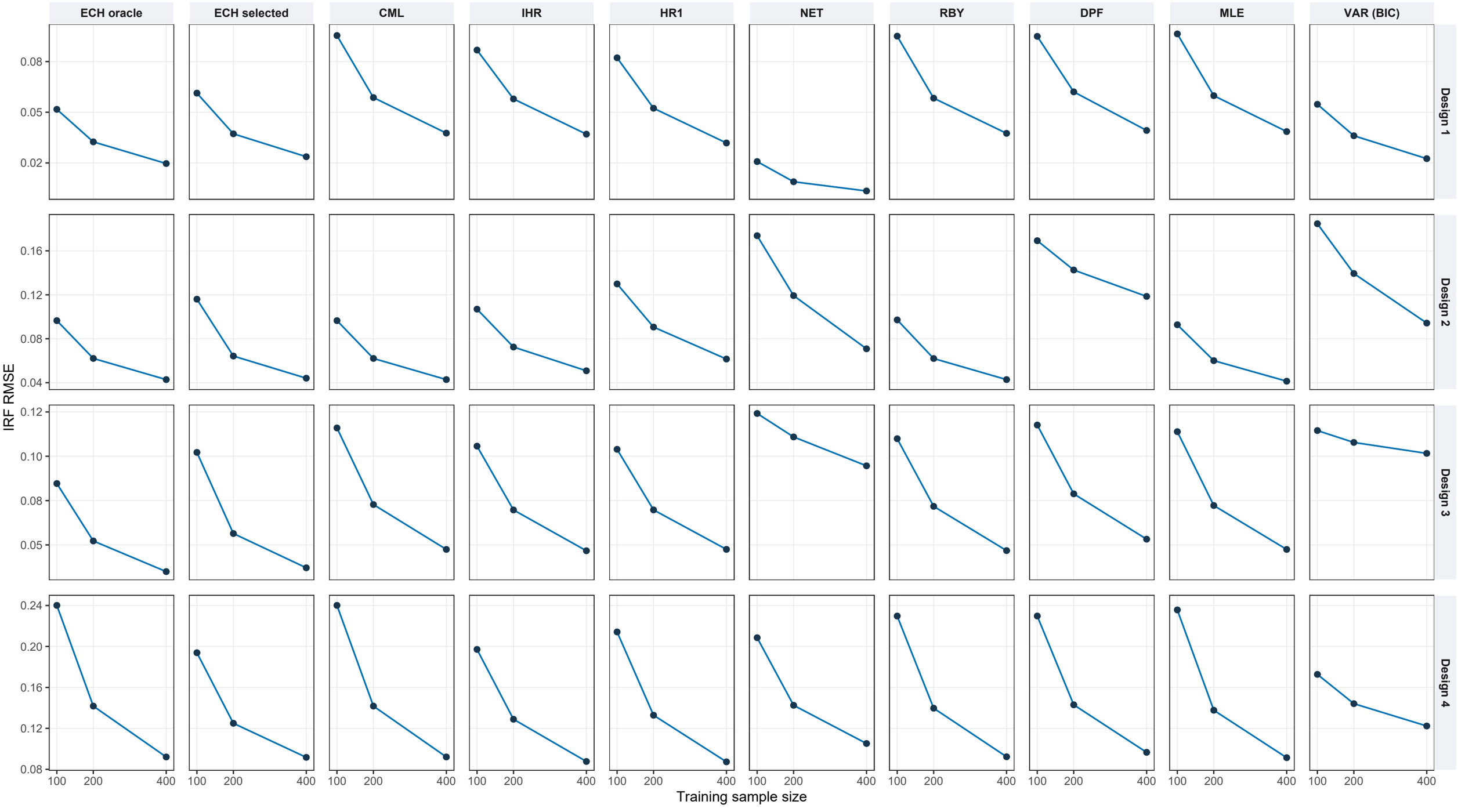
All available estimates | reduced-form IRFs, lags 1-5 | 1000 planned replications per cell



Square root of the mean replication loss after 1.0% trimming in each tail. This differs from ordinary RMSE.
Lines connect observed sample sizes; no convergence-rate curve is fitted.

Transfer-function accuracy

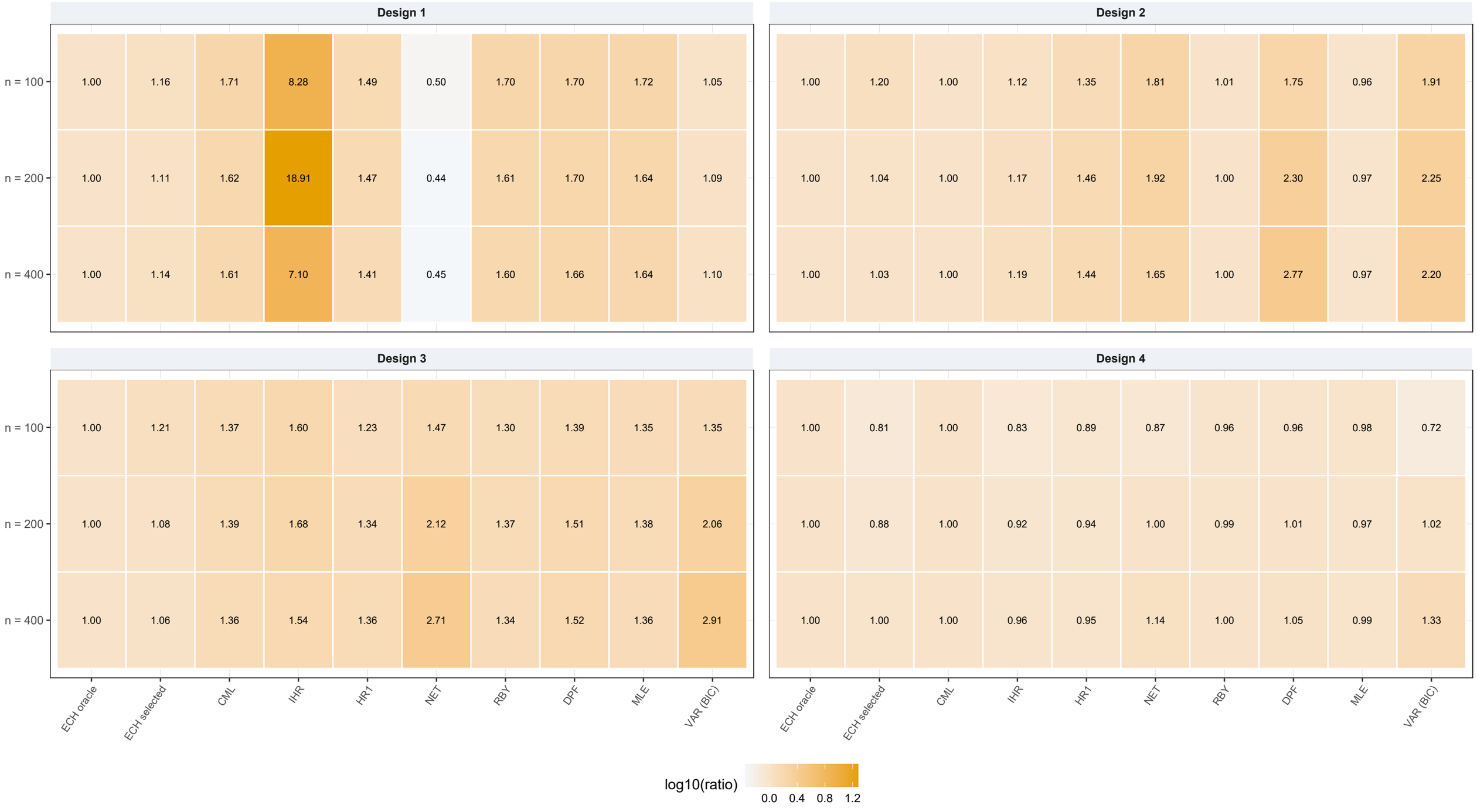
Qualified estimates | reduced-form IRFs, lags 1-5 | 1000 planned replications per cell



Untrimmed RMSE; whiskers show one Monte Carlo standard error (lower ends floored at zero).
Entries and lags are pooled within each replication. Lines do not impose a convergence rate. MCSEs can be unstable with extreme losses.

IRF RMSE relative to oracle echelon

All available estimates | each ratio uses common usable replications | 1 = equal RMSE



Numbers are RMSE ratios, not averages of replication-wise ratios. Colour represents log10(ratio).
Missing outputs remain in failure counts. Large finite errors are retained. See CSV tables for paired counts and Monte Carlo standard errors.

Forecast RMSE relative to the DGP forecast

All available estimates | each ratio uses common usable replications | 1 = equal RMSE

	Design 1										Design 2										Design 3										Design 4										
n = 100	1.01	1.01	1.03	1.03	1.02	1.00	1.03	1.03	1.03	1.01	1.03	1.04	1.03	1.06	1.11	1.14	1.03	1.14	1.03	1.14	1.04	1.05	1.07	1.07	1.06	1.06	1.06	511.38	1.08	1.05	1.11	1.09	1.11	1.08	1.07	1.14	1.11	16.30	1.12	1.05	h = 1
n = 200	1.00	1.01	1.01	1.01	1.01	1.00	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.03	1.06	1.07	1.01	1.10	1.01	1.09	1.02	1.03	1.05	1.04	1.04	1.08	1.05	1.03	1.05	1.07	1.05	1.02	1.05	1.03	1.02	1.06	1.05	1.06	1.04	1.03	
n = 400	1.00	1.01	1.01	1.01	1.01	1.00	1.01	1.01	1.01	1.00	1.00	1.00	1.00	1.02	1.04	1.03	1.00	1.07	1.00	1.06	1.01	1.01	1.01	1.02	1.01	1.04	1.01	1.01	1.01	1.04	1.01	1.01	1.01	1.01	1.01	1.03	1.01	1.02	1.02	1.03	
n = 100	1.00	1.01	1.02	1.04	1.02	1.00	1.02	1.02	1.02	1.01	1.01	1.02	1.01	1.02	1.04	1.05	1.01	1.05	1.01	1.06	1.03	1.03	1.07	1.06	1.04	1.04	1.05	567.68	1.07	1.03	1.10	1.07	1.10	1.07	1.07	1.10	1.10	20.81	1.09	1.04	h = 2
n = 200	1.00	1.01	1.01	1.02	1.01	1.00	1.01	1.01	1.01	1.00	1.00	1.00	1.00	1.01	1.02	1.02	1.00	1.03	1.00	1.04	1.02	1.01	1.02	1.04	1.02	1.04	1.02	1.03	1.03	1.04	1.05	1.02	1.05	1.03	1.03	1.04	1.04	1.05	1.04	1.02	
n = 400	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.01	1.01	1.00	1.02	1.00	1.03	1.01	1.01	1.02	1.02	1.01	1.03	1.02	1.02	1.02	1.04	1.02	1.01	1.02	1.01	1.01	1.03	1.02	1.02	1.02	1.02	
n = 100	1.00	1.00	1.01	1.45	1.00	1.00	1.01	1.01	1.01	1.00	1.02	1.02	1.02	1.02	1.03	1.04	1.02	1.03	1.01	1.03	1.02	1.02	1.04	1.13	1.03	1.02	1.04	593.37	1.05	1.02	1.09	1.05	1.09	1.07	1.06	1.09	1.10	23.83	1.10	1.04	h = 3
n = 200	1.00	1.00	1.01	1.20	1.00	1.00	1.01	1.00	1.01	1.00	1.01	1.01	1.01	1.01	1.02	1.02	1.01	1.02	1.01	1.02	1.01	1.01	1.02	1.14	1.01	1.02	1.02	1.02	1.02	1.02	1.04	1.02	1.04	1.03	1.02	1.04	1.04	1.05	1.04	1.02	
n = 400	1.00	1.00	1.00	1.04	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.01	1.01	1.00	1.01	1.00	1.02	1.01	1.01	1.00	1.10	1.01	1.02	1.00	1.01	1.01	1.03	1.02	1.02	1.02	1.02	1.02	1.03	1.02	1.03	1.02	1.02	
n = 100	1.00	1.00	1.01	4.91	1.00	1.00	1.01	1.00	1.01	1.00	1.01	1.02	1.01	1.01	1.02	1.03	1.01	1.02	1.01	1.03	1.01	1.01	1.03	1.50	1.01	1.02	1.02	664.23	1.03	1.02	1.09	1.06	1.09	1.09	1.06	1.09	1.09	21.57	1.10	1.05	h = 4
n = 200	1.00	1.00	1.00	4.05	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.01	1.01	1.01	1.02	1.01	1.01	1.02	1.01	1.02	1.00	1.00	1.02	1.83	1.01	1.01	1.02	1.01	1.02	1.02	1.05	1.02	1.05	1.07	1.02	1.03	1.05	1.05	1.04	1.03	
n = 400	1.00	1.00	1.00	1.58	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.00	1.00	1.01	1.00	1.01	1.00	1.00	1.01	1.84	1.00	1.01	1.01	1.00	1.01	1.01	1.02	1.02	1.02	1.03	1.02	1.03	1.02	1.03	1.02	1.03	
n = 100	1.00	1.00	1.00	23.07	1.00	1.00	1.00	1.00	1.00	1.00	1.02	1.02	1.02	1.02	1.02	1.03	1.02	1.02	1.02	1.02	1.01	1.01	1.02	2.94	1.01	1.00	1.03	759.31	1.03	1.00	1.09	1.06	1.09	1.14	1.06	1.09	1.09	23.48	1.09	1.04	h = 5
n = 200	1.00	1.00	1.00	23.37	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.00	1.00	1.01	4.84	1.00	1.01	1.01	1.01	1.01	1.01	1.04	1.02	1.04	1.28	1.02	1.02	1.04	1.04	1.04	1.02	
n = 400	1.00	1.00	1.00	6.07	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.00	1.00	1.01	1.00	1.01	1.00	1.00	1.00	6.26	1.00	1.01	1.00	1.01	1.01	1.01	1.02	1.02	1.02	1.04	1.01	1.02	1.02	1.03	1.02	1.02	
	ECH oracle	ECH selected	CML	IHR	HR1	NET	RBY	DPF	MLE	VAR (BIC)	ECH oracle	ECH selected	CML	IHR	HR1	NET	RBY	DPF	MLE	VAR (BIC)	ECH oracle	ECH selected	CML	IHR	HR1	NET	RBY	DPF	MLE	VAR (BIC)	ECH oracle	ECH selected	CML	IHR	HR1	NET	RBY	DPF	MLE	VAR (BIC)		

log10(ratio)

0 1 2

Numbers are RMSE ratios, not averages of replication-wise ratios. Colour represents log10(ratio).
Missing outputs remain in failure counts. Large finite errors are retained. See CSV tables for paired counts and Monte Carlo standard errors.

Forecast RMSE relative to the DGP forecast

Qualified estimates | each ratio uses common usable replications | 1 = equal RMSE

	Design 1										Design 2									
n = 100	1.01	1.01	1.03	1.03	1.02	1.00	1.03	1.03	1.03	1.01	1.03	1.04	1.03	1.06	1.11	1.14	1.03	1.14	1.03	1.14
n = 200	1.00	1.01	1.01	1.01	1.01	1.00	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.03	1.06	1.07	1.01	1.10	1.01	1.09
n = 400	1.00	1.01	1.01	1.01	1.01	1.00	1.01	1.01	1.01	1.00	1.00	1.00	1.00	1.02	1.04	1.03	1.00	1.07	1.00	1.06
n = 100	1.00	1.01	1.02	1.02	1.02	1.00	1.02	1.02	1.02	1.01	1.01	1.02	1.01	1.02	1.04	1.05	1.01	1.05	1.01	1.06
n = 200	1.00	1.01	1.01	1.01	1.01	1.00	1.01	1.01	1.01	1.00	1.00	1.00	1.00	1.01	1.02	1.02	1.00	1.03	1.00	1.04
n = 400	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.01	1.01	1.00	1.02	1.00	1.03
n = 100	1.00	1.00	1.01	1.01	1.00	1.00	1.01	1.01	1.01	1.00	1.02	1.02	1.02	1.02	1.03	1.04	1.02	1.03	1.01	1.03
n = 200	1.00	1.00	1.01	1.00	1.00	1.00	1.01	1.00	1.01	1.00	1.01	1.01	1.01	1.01	1.02	1.02	1.01	1.02	1.01	1.02
n = 400	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.01	1.01	1.00	1.01	1.00	1.02
n = 100	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.02	1.01	1.01	1.02	1.03	1.01	1.02	1.01	1.03
n = 200	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.01	1.01	1.01	1.02	1.01	1.01	1.02	1.01	1.02
n = 400	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.01	1.01	1.00	1.01	1.00	1.02

	Design 3										Design 4									
n = 100	1.04	1.05	1.07	1.06	1.06	1.06	1.06	1.08	1.08	1.05	1.11	1.09	1.11	1.08	1.07	1.15	1.11	1.15	1.12	1.05
n = 200	1.02	1.03	1.05	1.04	1.04	1.08	1.05	1.03	1.05	1.07	1.05	1.02	1.05	1.03	1.02	1.06	1.05	1.06	1.04	1.03
n = 400	1.01	1.01	1.01	1.01	1.01	1.04	1.01	1.01	1.01	1.04	1.01	1.01	1.01	1.01	1.01	1.03	1.01	1.02	1.02	1.03
n = 100	1.03	1.03	1.07	1.04	1.04	1.04	1.05	1.06	1.07	1.03	1.10	1.07	1.10	1.07	1.06	1.10	1.10	1.14	1.09	1.04
n = 200	1.02	1.01	1.02	1.02	1.02	1.04	1.02	1.03	1.03	1.04	1.05	1.02	1.05	1.03	1.03	1.03	1.04	1.05	1.04	1.02
n = 400	1.01	1.01	1.02	1.01	1.01	1.03	1.02	1.02	1.02	1.04	1.02	1.01	1.02	1.01	1.01	1.03	1.02	1.02	1.02	1.02
n = 100	1.02	1.02	1.04	1.04	1.03	1.02	1.04	1.04	1.05	1.02	1.09	1.05	1.09	1.06	1.06	1.10	1.10	1.14	1.10	1.04
n = 200	1.01	1.01	1.02	1.01	1.01	1.02	1.02	1.02	1.02	1.02	1.04	1.02	1.04	1.03	1.02	1.03	1.04	1.05	1.04	1.02
n = 400	1.01	1.01	1.00	1.01	1.01	1.02	1.00	1.01	1.01	1.03	1.02	1.01	1.02	1.02	1.01	1.02	1.02	1.02	1.02	1.02
n = 100	1.01	1.01	1.03	1.01	1.01	1.01	1.02	1.03	1.03	1.02	1.09	1.06	1.09	1.07	1.06	1.09	1.09	1.13	1.10	1.05
n = 200	1.00	1.00	1.02	1.01	1.01	1.01	1.02	1.01	1.02	1.02	1.05	1.02	1.05	1.03	1.02	1.03	1.05	1.05	1.04	1.03
n = 400	1.00	1.00	1.01	1.01	1.00	1.01	1.01	1.00	1.01	1.01	1.02	1.02	1.02	1.02	1.02	1.03	1.02	1.03	1.02	1.03
n = 100	1.01	1.01	1.03	1.01	1.01	1.01	1.03	1.01	1.03	1.00	1.09	1.06	1.09	1.07	1.06	1.09	1.09	1.14	1.09	1.04
n = 200	1.00	1.00	1.01	1.01	1.00	1.01	1.01	1.01	1.01	1.01	1.04	1.02	1.04	1.03	1.02	1.02	1.04	1.04	1.04	1.02
n = 400	1.00	1.00	1.00	1.00	1.00	1.01	1.00	1.01	1.01	1.01	1.02	1.02	1.02	1.02	1.01	1.02	1.02	1.03	1.02	1.02

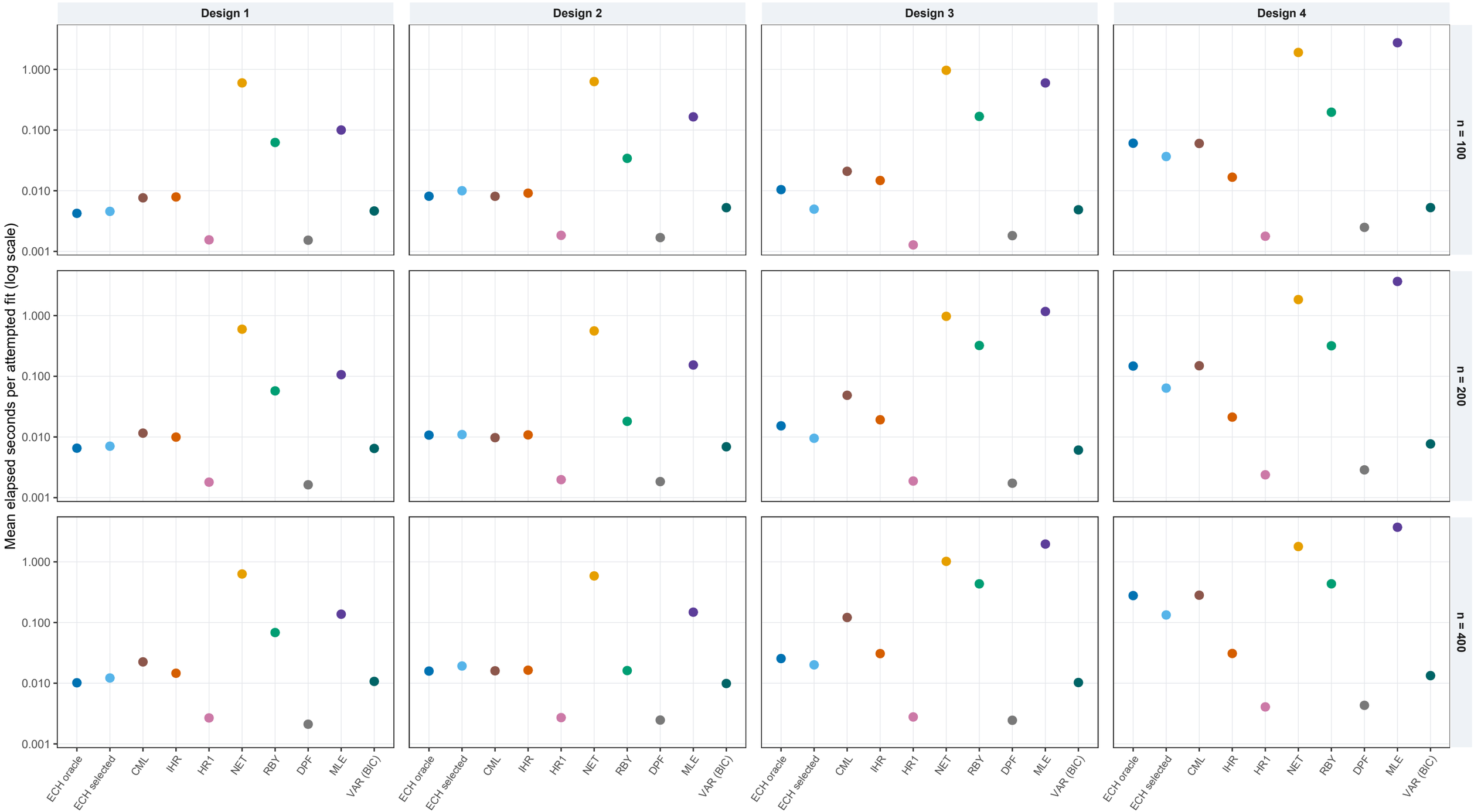
ECH oracle · ECH selected · CML · IHR · HR1 · NET · RBY · DPF · MLE · VAR (BIC)

log10(ratio)
0.00 0.02 0.04 0.06

Numbers are RMSE ratios, not averages of replication-wise ratios. Colour represents log10(ratio).
Missing outputs remain in failure counts. Large finite errors are retained. See CSV tables for paired counts and Monte Carlo standard errors.

Estimation time

Elapsed times measured inside simulation workers, including failed attempts



Points show mean per-fit elapsed times under the recorded parallel workload. Summing worker times does not give wall-clock duration.
DGP filtering, forecast evaluation and checkpoint overhead are excluded. See timing.csv for medians, 90th percentiles and total worker times.

Numerical diagnostics

Percent of attempted replications; categories can overlap

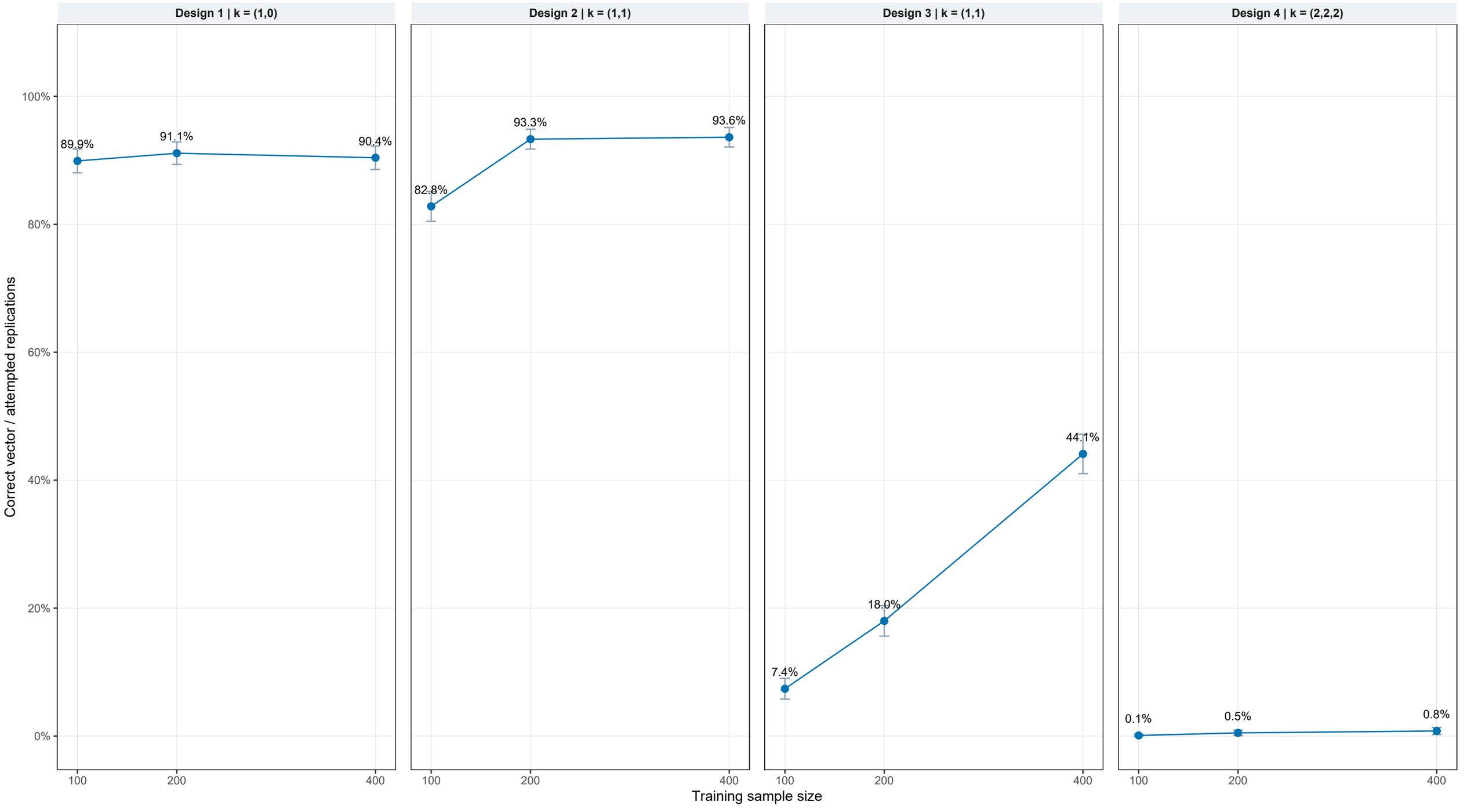
	Design 1										Design 2										Design 3										Design 4									
Nonconverged																																								
n = 100	0.0	0.0	0.0	37.4	0.0	37.4	1.7	0.0	0.0	0.0	0.0	0.0	0.0	19.9	0.0	19.9	0.1	0.0	0.0	0.0	0.0	0.0	0.0	96.4	0.0	96.4	3.0	0.0	0.0	0.0	0.0	0.0	0.0	81.0	0.0	81.0	3.0	0.0	0.0	0.0
n = 200	0.0	0.0	0.0	37.8	0.0	37.8	0.7	0.0	0.0	0.0	0.0	0.0	0.0	21.0	0.0	21.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	97.1	0.0	97.1	6.9	0.0	0.0	0.0	0.0	0.0	0.0	67.9	0.0	67.9	4.6	0.0	0.0	0.0
n = 400	0.0	0.0	0.0	34.5	0.0	34.5	0.8	0.0	0.0	0.0	0.0	0.0	0.0	25.5	0.0	25.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	95.2	0.0	95.2	8.3	0.0	0.1	0.0	0.0	0.0	0.0	51.3	0.0	51.3	4.7	0.0	0.0	0.0
Forecast unavailable																																								
n = 100	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
n = 200	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
n = 400	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AR unstable																																								
n = 100	0.0	0.0	0.0	7.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	21.8	0.0	0.3	0.0	0.8	0.0	0.0	0.0	0.0	0.0	8.3	0.2	3.4	0.0	0.9	0.0	0.1
n = 200	0.0	0.0	0.0	8.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	19.6	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	7.1	0.0	5.6	0.0	0.0	0.0	0.0
n = 400	0.0	0.0	0.0	9.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	22.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.4	0.0	8.8	0.0	0.0	0.0	0.0
MA noninvertible																																								
n = 100	0.0	0.0	0.0	8.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.1	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	30.7	2.1	0.5	0.0	0.9	0.0	0.0	0.0	0.0	0.0	11.4	0.1	10.0	0.0	0.3	0.0	0.0
n = 200	0.0	0.0	0.0	9.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	27.4	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	8.0	0.0	16.4	0.0	0.0	0.0	0.0
n = 400	0.0	0.0	0.0	9.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	25.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.6	0.0	25.8	0.0	0.0	0.0	0.0
Qualified																																								
n = 100	100.0	100.0	100.0	91.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	99.4	99.9	99.3	100.0	100.0	100.0	100.0	100.0	100.0	100.0	68.4	97.9	99.4	100.0	98.9	100.0	100.0	100.0	100.0	100.0	88.0	99.7	89.4	100.0	98.9	100.0	99.9
n = 200	100.0	100.0	100.0	90.5	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	72.2	99.8	99.8	100.0	99.9	100.0	100.0	100.0	100.0	100.0	91.6	100.0	83.3	100.0	100.0	100.0	100.0
n = 400	100.0	100.0	100.0	90.3	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	73.9	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	93.3	100.0	74.0	100.0	100.0	100.0	100.0
	ECH oracle	ECH selected	CML	IHR	HR1	NET	RBY	DPF	MLE	VAR (BIC)	ECH oracle	ECH selected	CML	IHR	HR1	NET	RBY	DPF	MLE	VAR (BIC)	ECH oracle	ECH selected	CML	IHR	HR1	NET	RBY	DPF	MLE	VAR (BIC)	ECH oracle	ECH selected	CML	IHR	HR1	NET	RBY	DPF	MLE	VAR (BIC)

Percent: 0 25 50 75 100

Qualified requires stable AR, invertible MA and reported convergence when applicable. NET's convergence flag refers to its IHR pilot.
HR1, DPF and VAR (BIC) have no iterative convergence requirement. Forecast availability also requires finite squared errors. These are diagnostics, not proof of a global optimum.

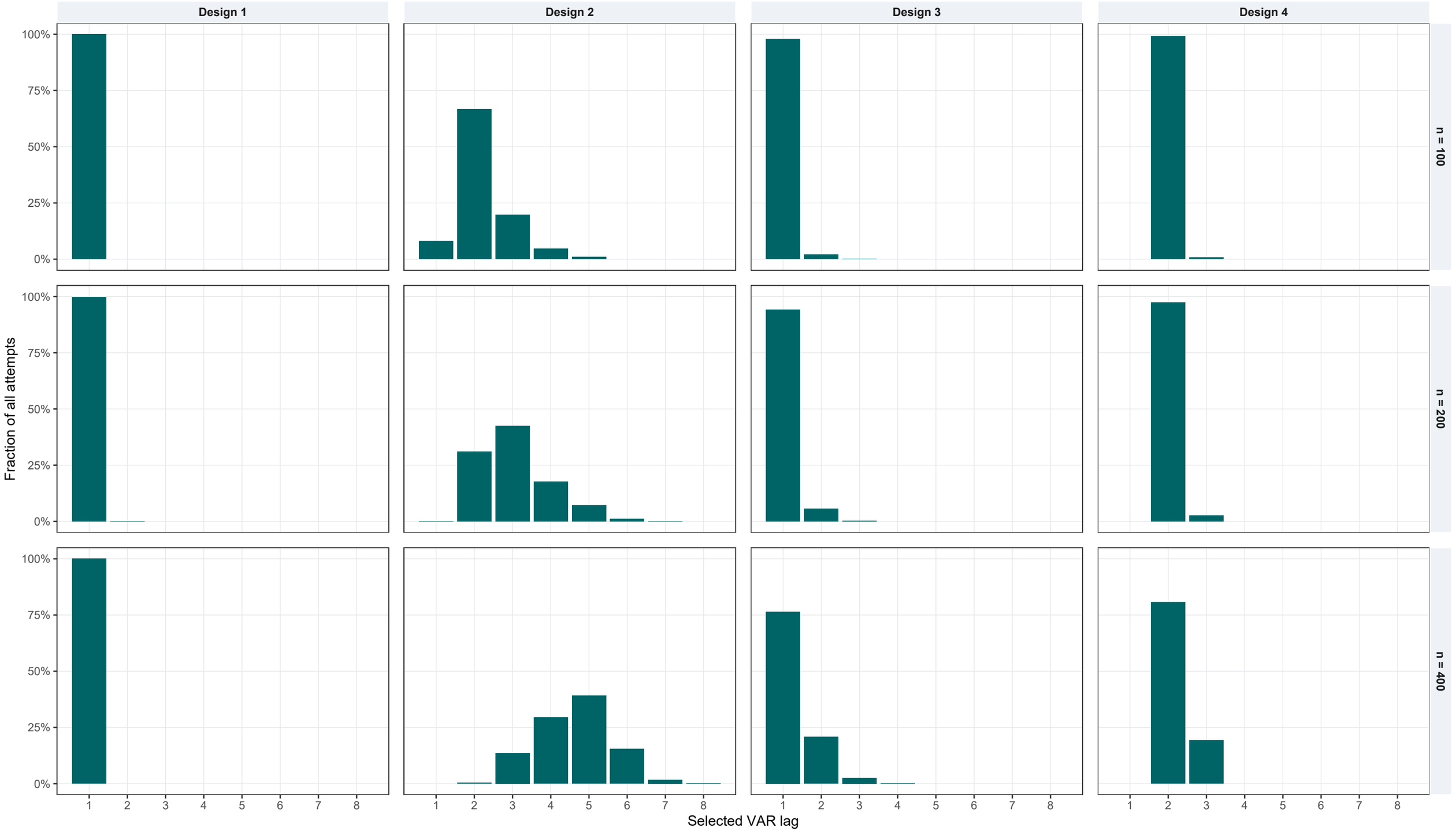
Data-driven Kronecker index selection

Exact recovery of the population index vector



Whiskers: +/-1.96 binomial Monte Carlo standard errors (clipped to [0,1]). Unreturned indices remain in the denominator.
An index vector may be unavailable because the subsequent fit failed; this is not a pure selection-error probability. Full vector frequencies are in indices.csv.

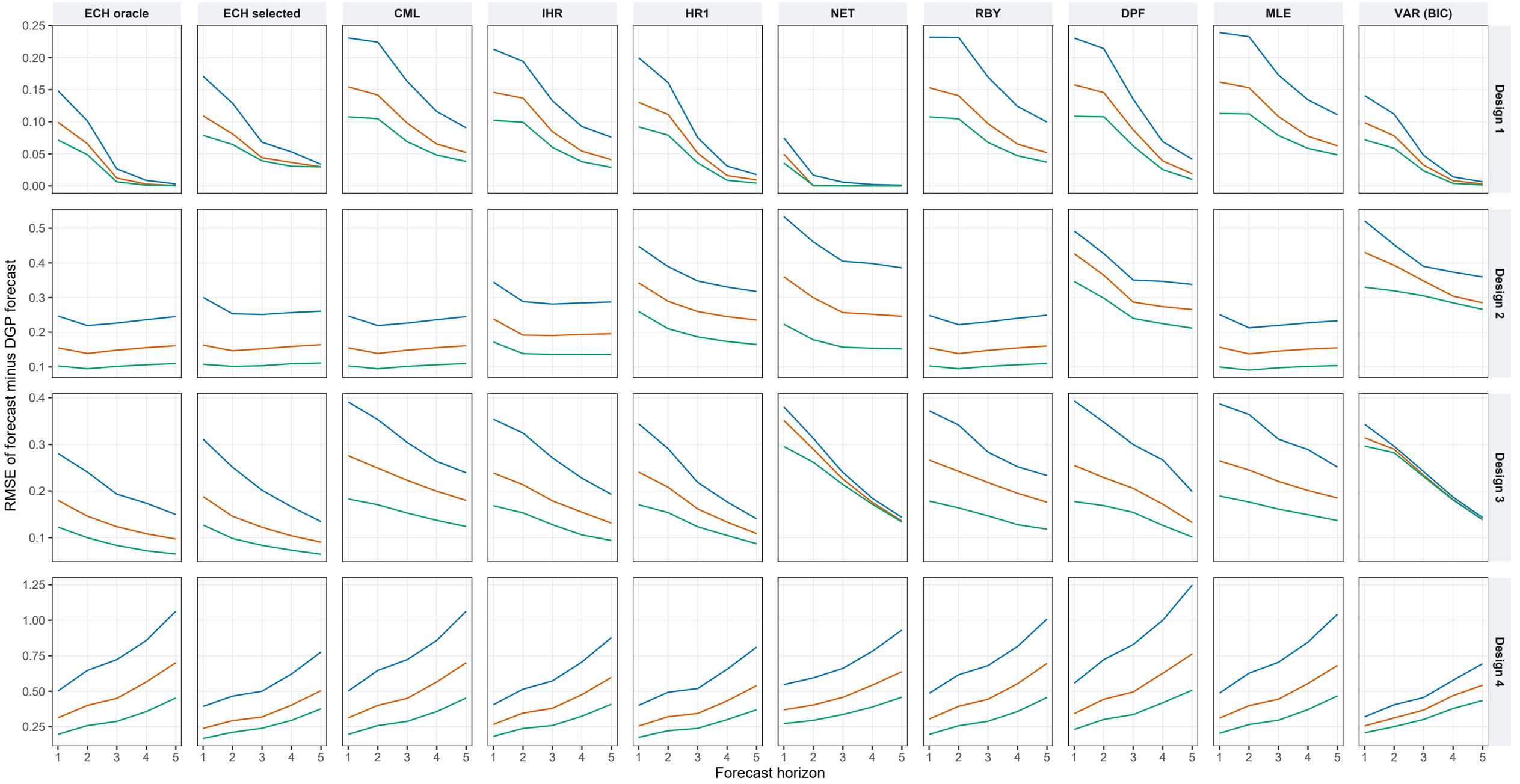
VAR lag selection by BIC



BIC compares candidates on a common sample; the selected order is refitted on all available rows. Failed fits remain in the denominator.
These are approximation orders, not recovery probabilities for the VARMA population's AR order.

Forecast estimation error

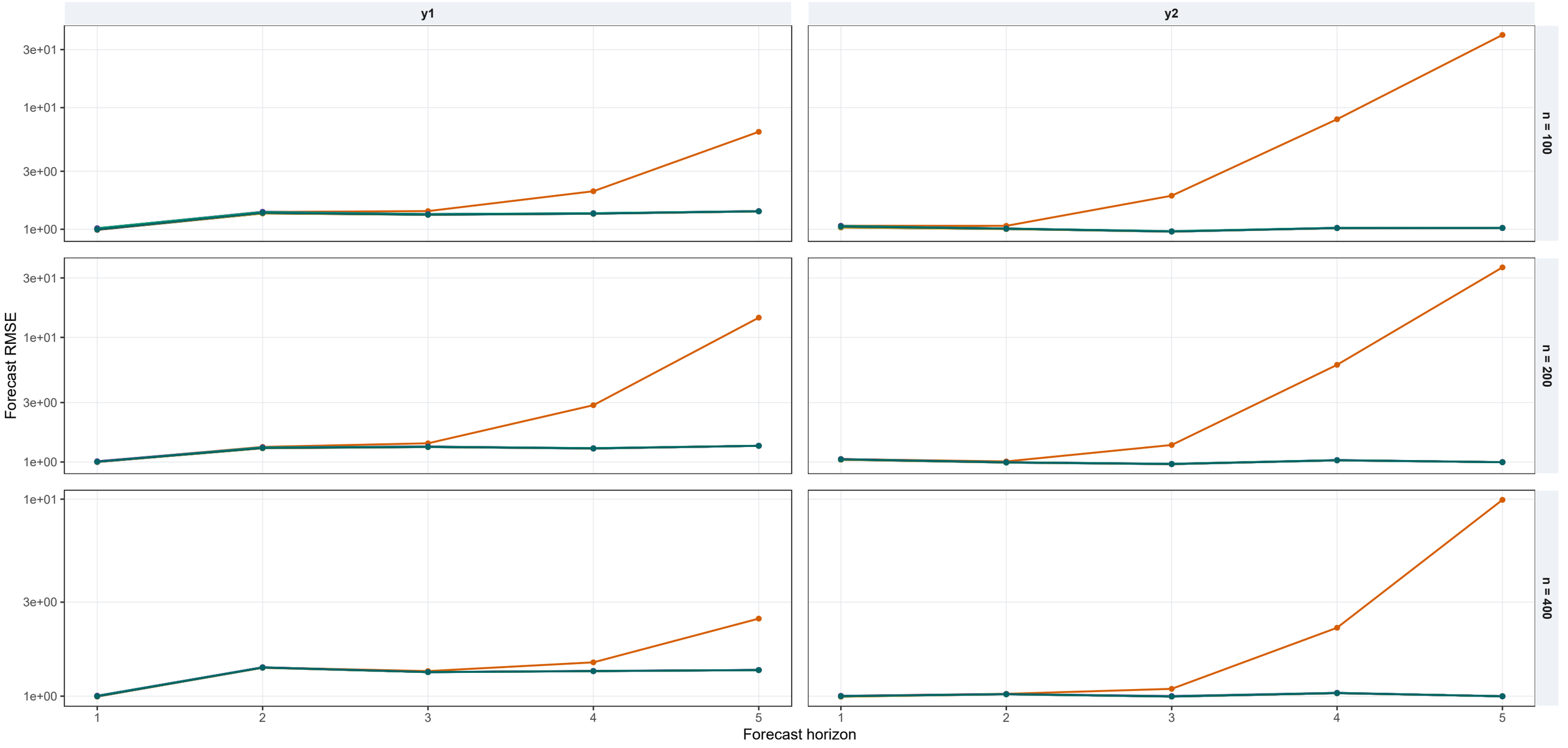
Qualified estimates | variables pooled within each replication | untrimmed



This measures distance from the true-parameter conditional forecast using the same observed history.
It separates forecast estimation error from unpredictable future innovations. Available-subset values, including extreme errors, are retained in forecast_gap.csv.

Forecast accuracy: Design 1

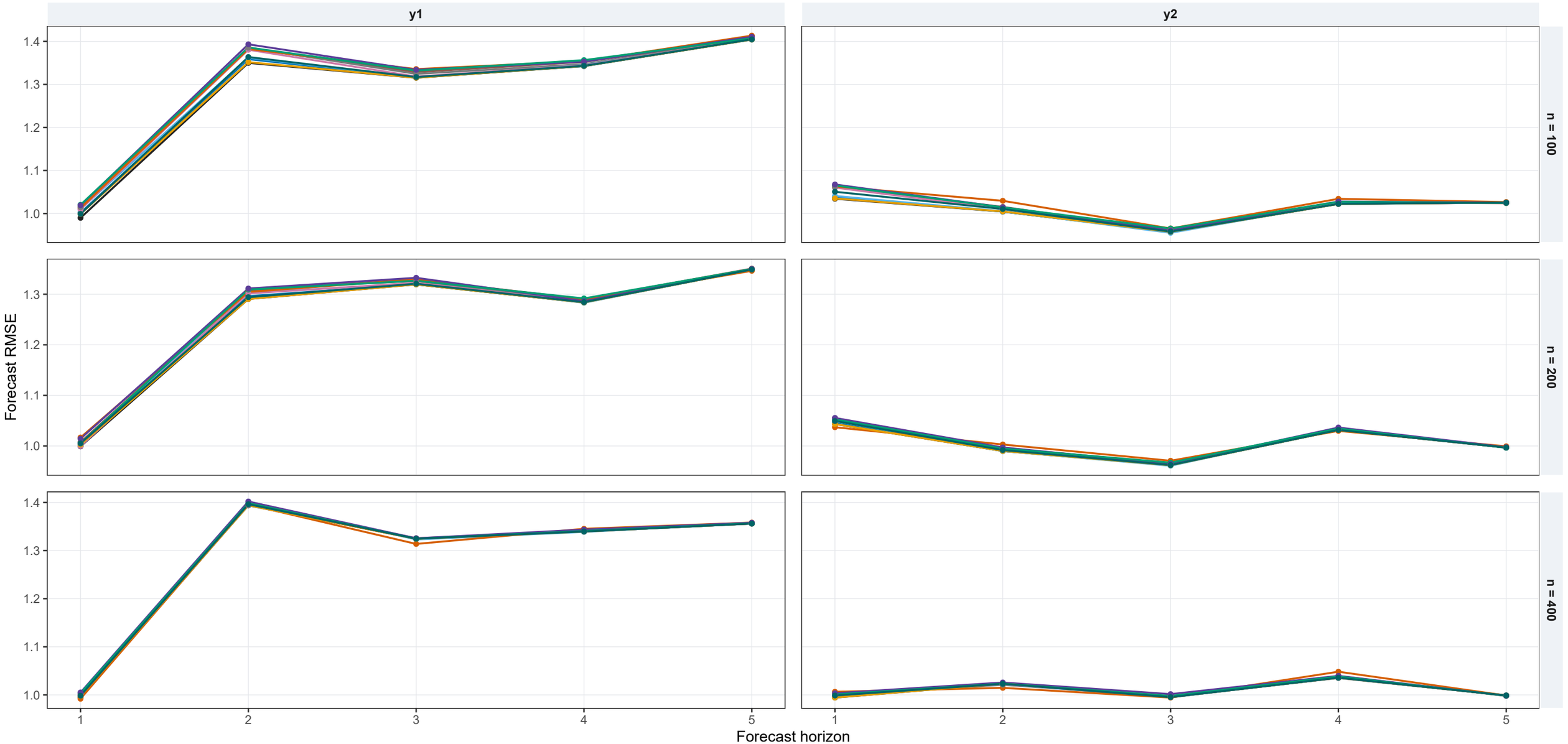
All available estimates | logarithmic vertical axis | untrimmed RMSE



Errors are realized future observations minus forecasts. DGP uses true parameters and the same observed training history.
Qualified means stable AR, invertible MA and reported convergence where applicable; NET uses its IHR pilot's flag. Counts are in the diagnostic tables.

Forecast accuracy: Design 1

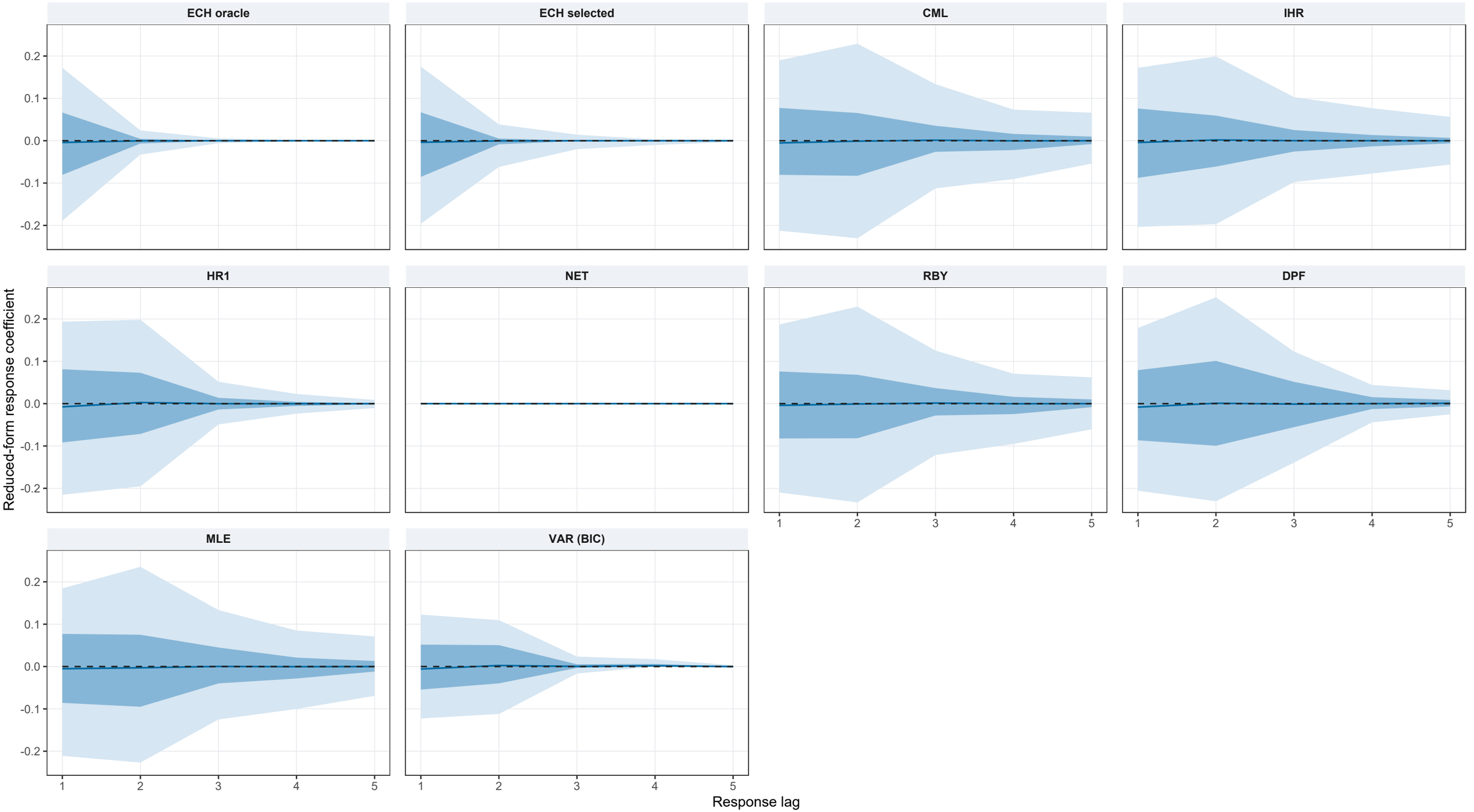
Qualified estimates | linear vertical axis | untrimmed RMSE



Errors are realized future observations minus forecasts. DGP uses true parameters and the same observed training history.
Qualified means stable AR, invertible MA and reported convergence where applicable; NET uses its IHR pilot's flag. Counts are in the diagnostic tables.

Design 1: response y1 to innovation 1

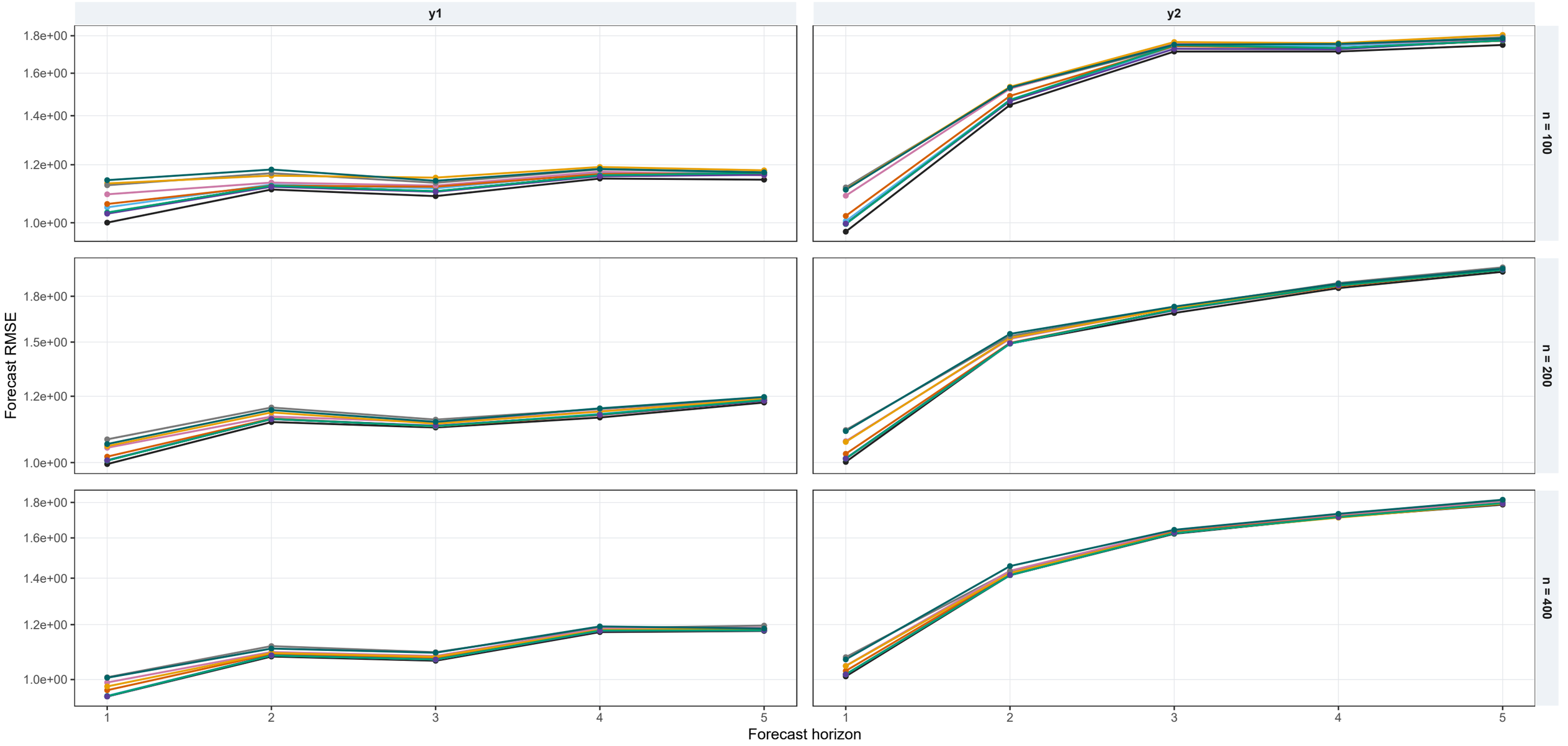
Sampling distribution of IRF estimates | n = 100



Blue line: median. Dark band: 25th-75th percentiles. Light band: 5th-95th percentiles. Dashed line: population response.
Bands describe variation across Monte Carlo replications; they are not confidence intervals. Extreme losses remain in the RMSE summaries.

Forecast accuracy: Design 2

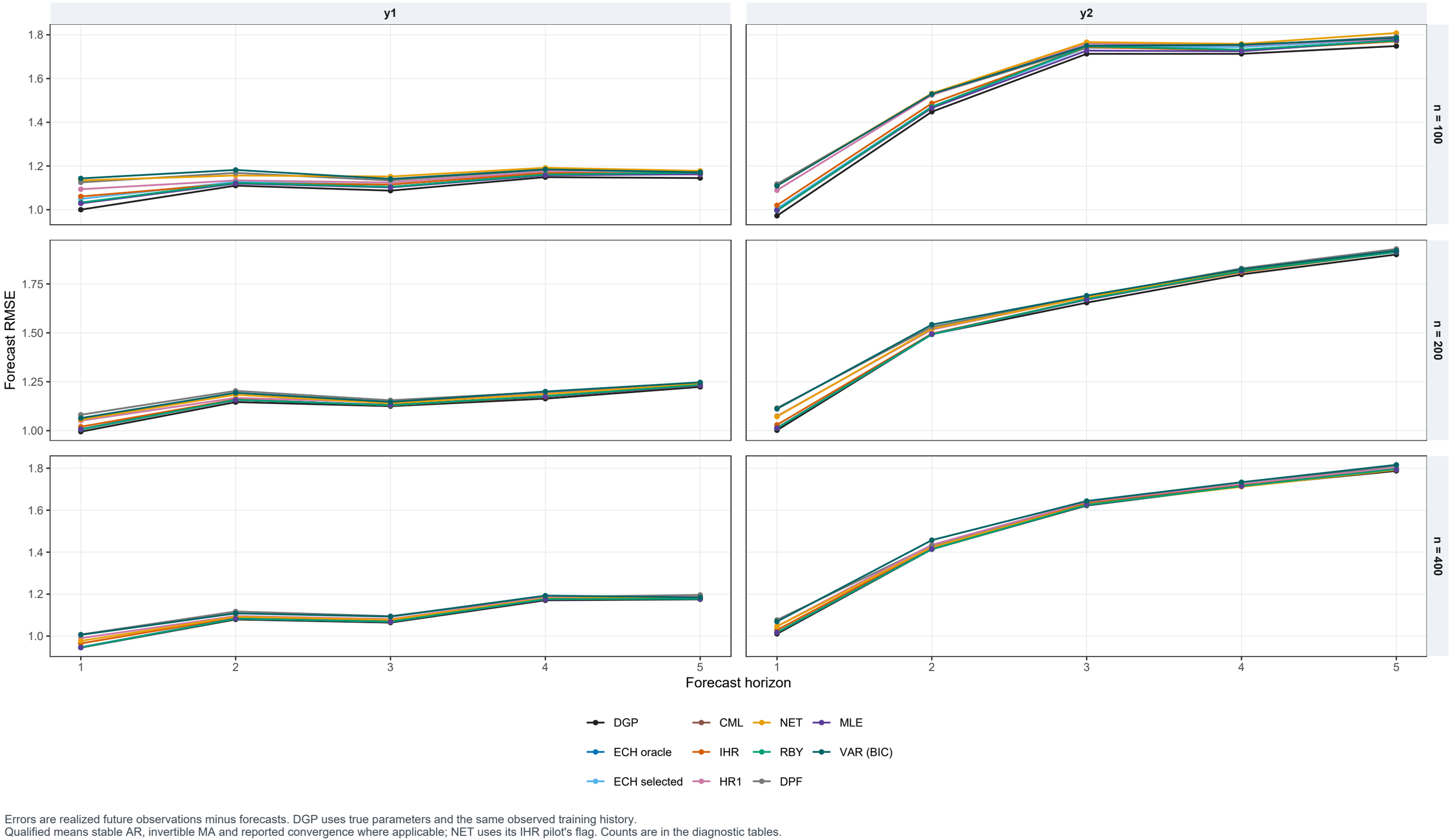
All available estimates | logarithmic vertical axis | untrimmed RMSE



Errors are realized future observations minus forecasts. DGP uses true parameters and the same observed training history.
Qualified means stable AR, invertible MA and reported convergence where applicable; NET uses its IHR pilot's flag. Counts are in the diagnostic tables.

Forecast accuracy: Design 2

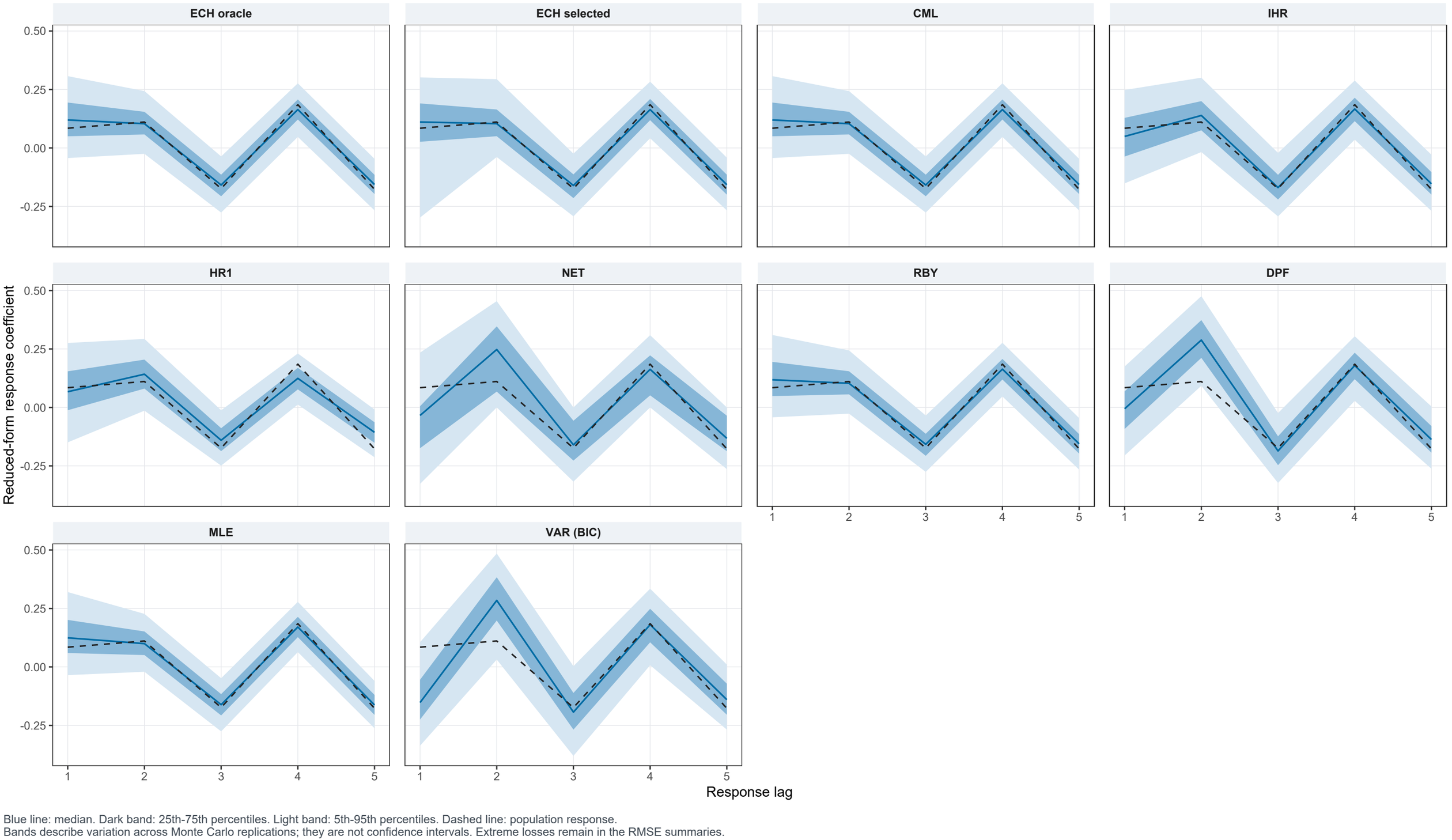
Qualified estimates | linear vertical axis | untrimmed RMSE



Errors are realized future observations minus forecasts. DGP uses true parameters and the same observed training history.
Qualified means stable AR, invertible MA and reported convergence where applicable; NET uses its IHR pilot's flag. Counts are in the diagnostic tables.

Design 2: response y1 to innovation 1

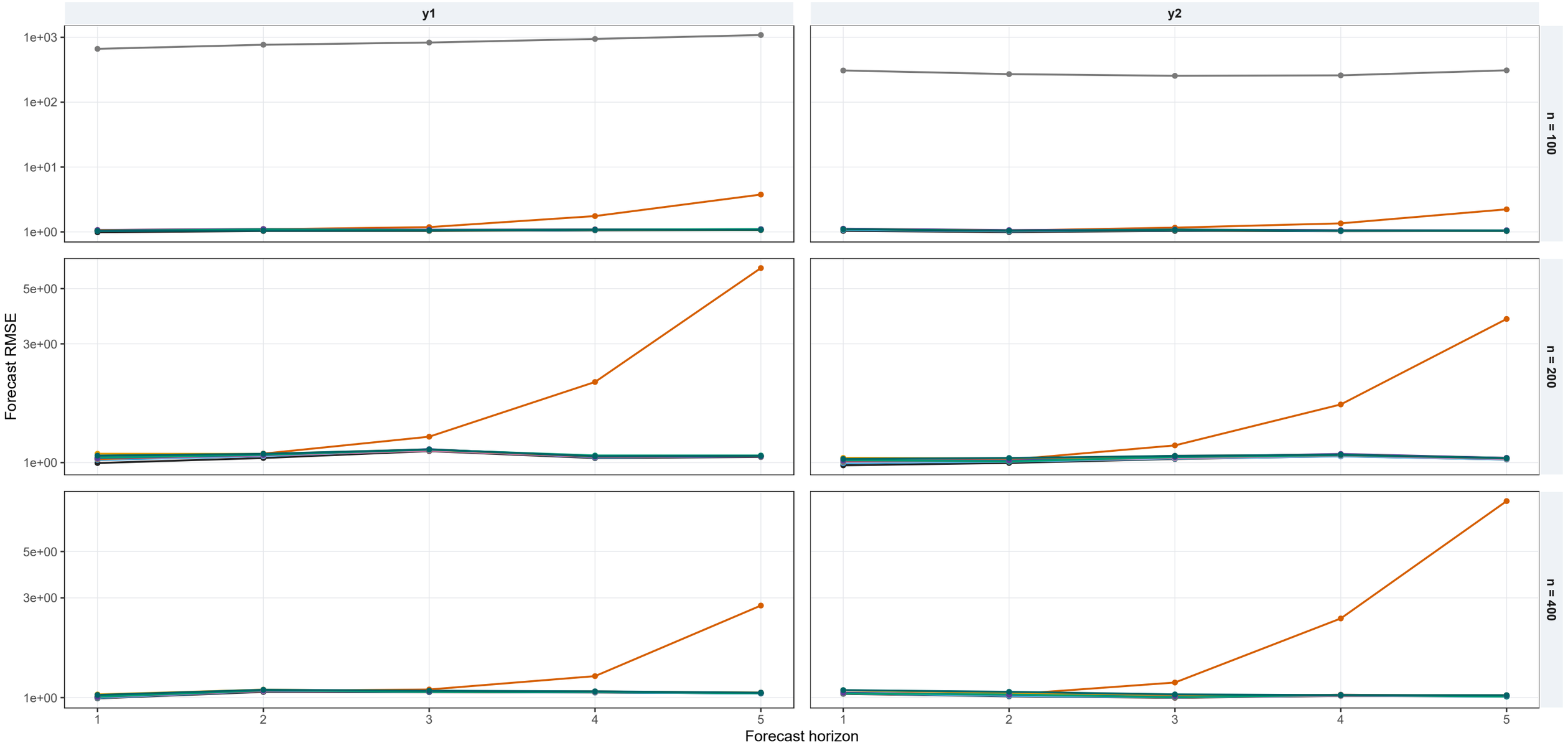
Sampling distribution of IRF estimates | n = 100



Blue line: median. Dark band: 25th-75th percentiles. Light band: 5th-95th percentiles. Dashed line: population response.
Bands describe variation across Monte Carlo replications; they are not confidence intervals. Extreme losses remain in the RMSE summaries.

Forecast accuracy: Design 3

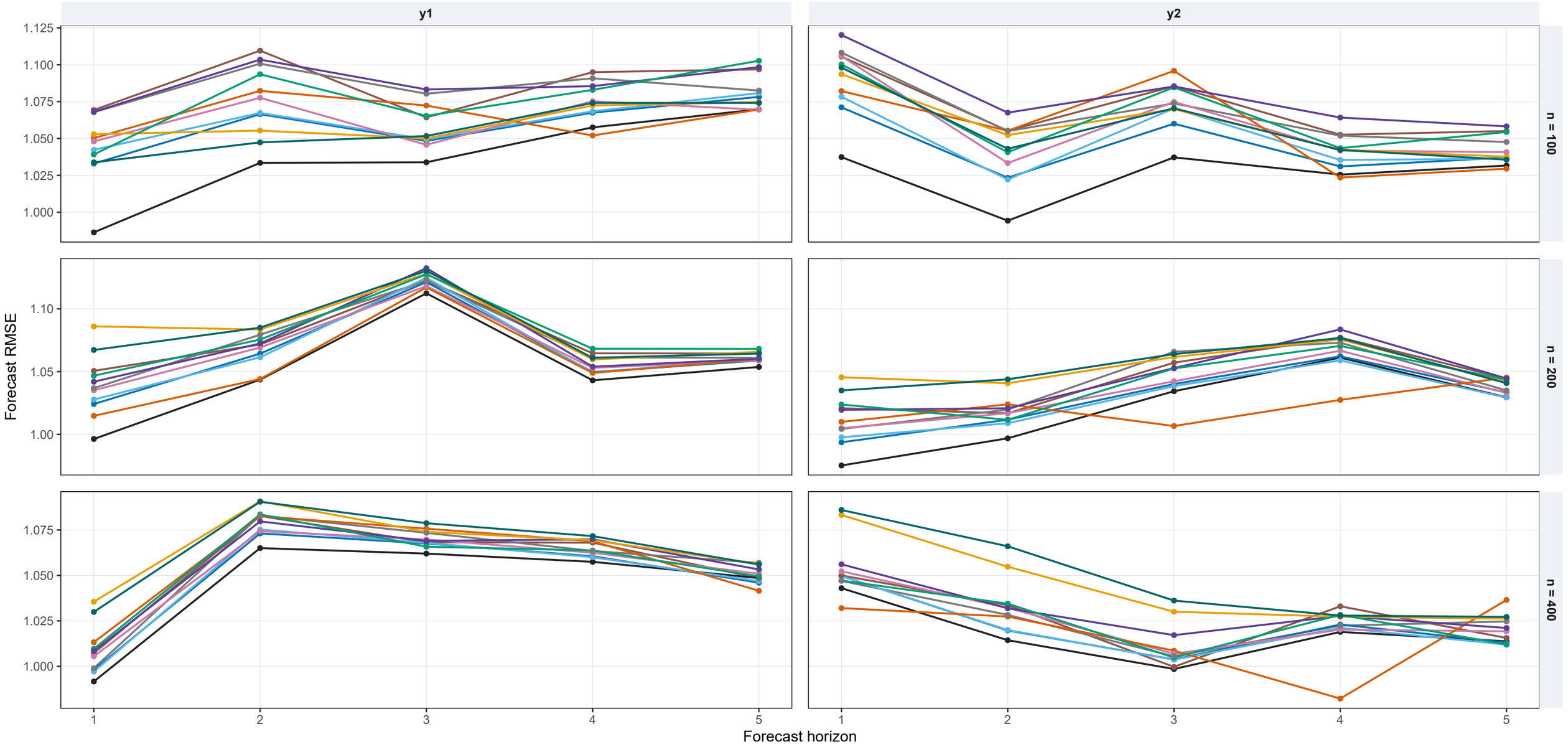
All available estimates | logarithmic vertical axis | untrimmed RMSE



Errors are realized future observations minus forecasts. DGP uses true parameters and the same observed training history.
Qualified means stable AR, invertible MA and reported convergence where applicable; NET uses its IHR pilot's flag. Counts are in the diagnostic tables.

Forecast accuracy: Design 3

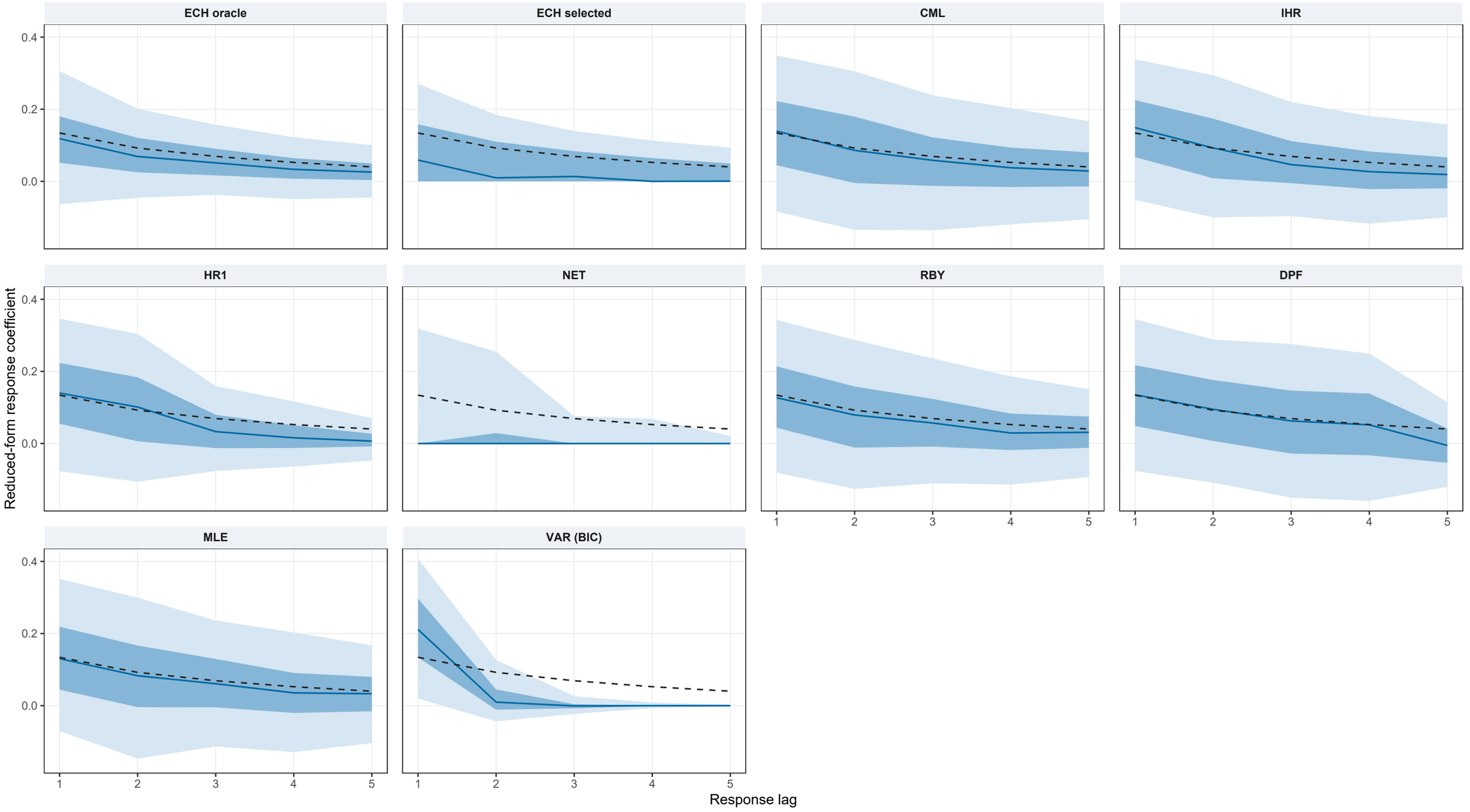
Qualified estimates | linear vertical axis | untrimmed RMSE



Errors are realized future observations minus forecasts. DGP uses true parameters and the same observed training history.
Qualified means stable AR, invertible MA and reported convergence where applicable; NET uses its IHR pilot's flag. Counts are in the diagnostic tables.

Design 3: response y1 to innovation 1

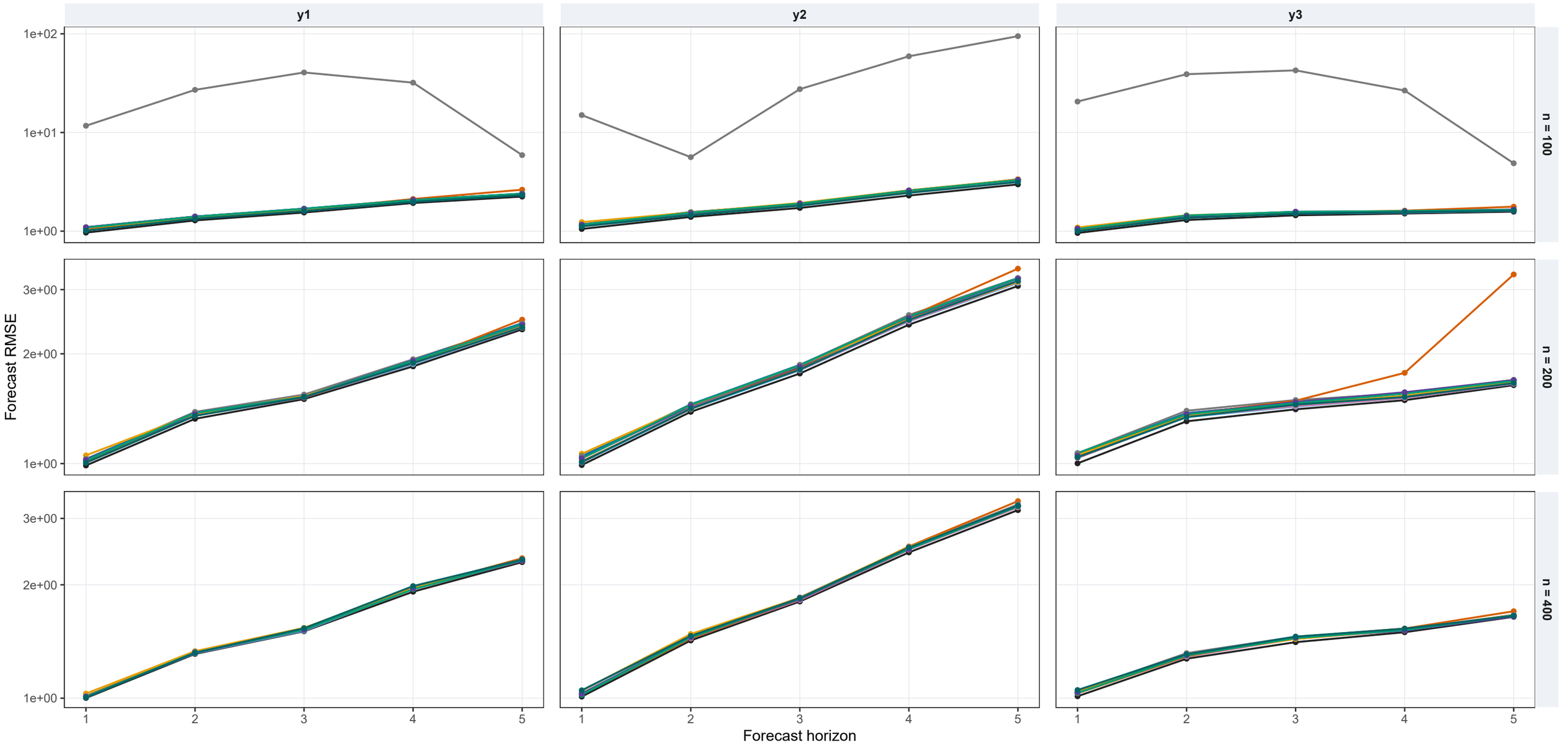
Sampling distribution of IRF estimates | n = 100



Blue line: median. Dark band: 25th-75th percentiles. Light band: 5th-95th percentiles. Dashed line: population response.
Bands describe variation across Monte Carlo replications; they are not confidence intervals. Extreme losses remain in the RMSE summaries.

Forecast accuracy: Design 4

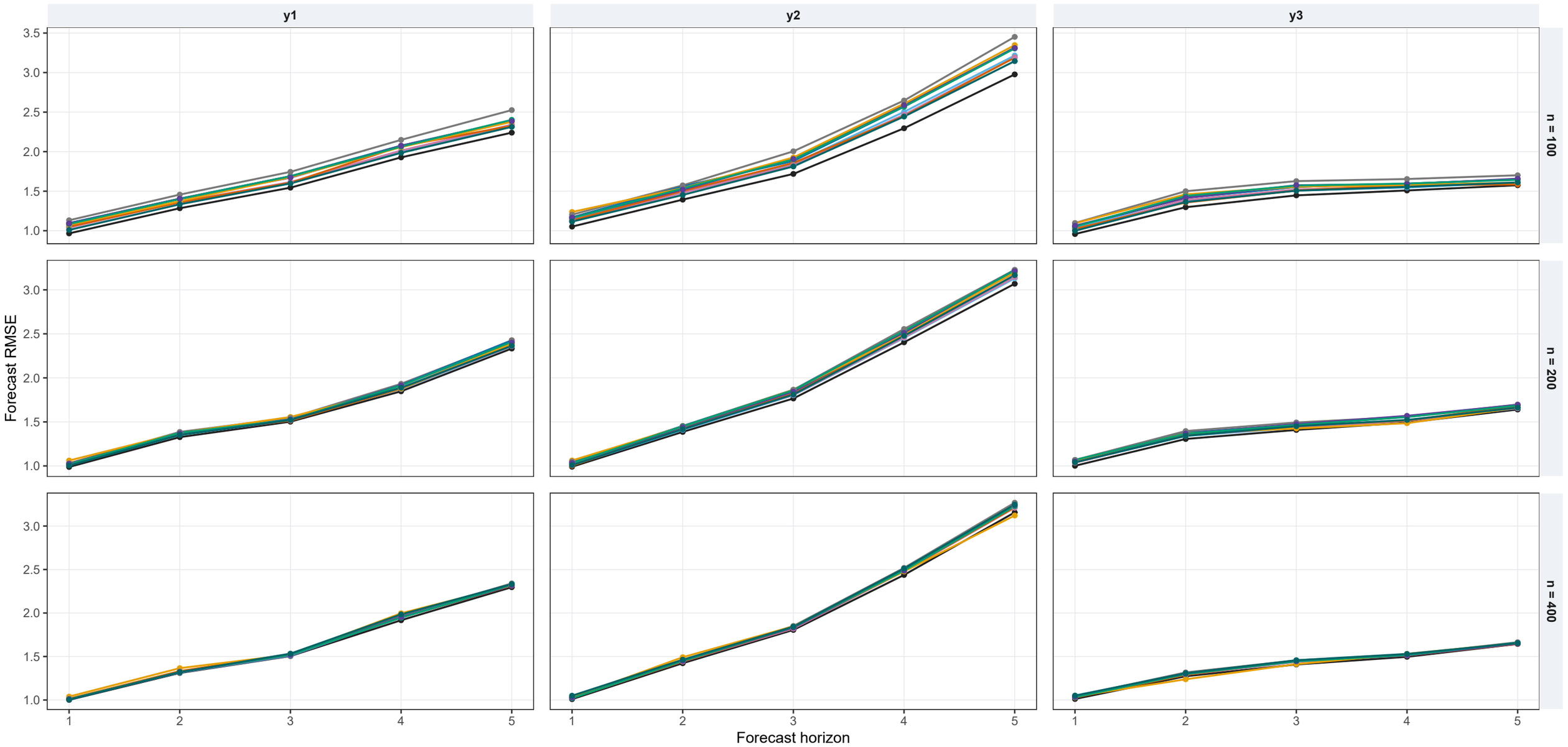
All available estimates | logarithmic vertical axis | untrimmed RMSE



Errors are realized future observations minus forecasts. DGP uses true parameters and the same observed training history.
Qualified means stable AR, invertible MA and reported convergence where applicable; NET uses its IHR pilot's flag. Counts are in the diagnostic tables.

Forecast accuracy: Design 4

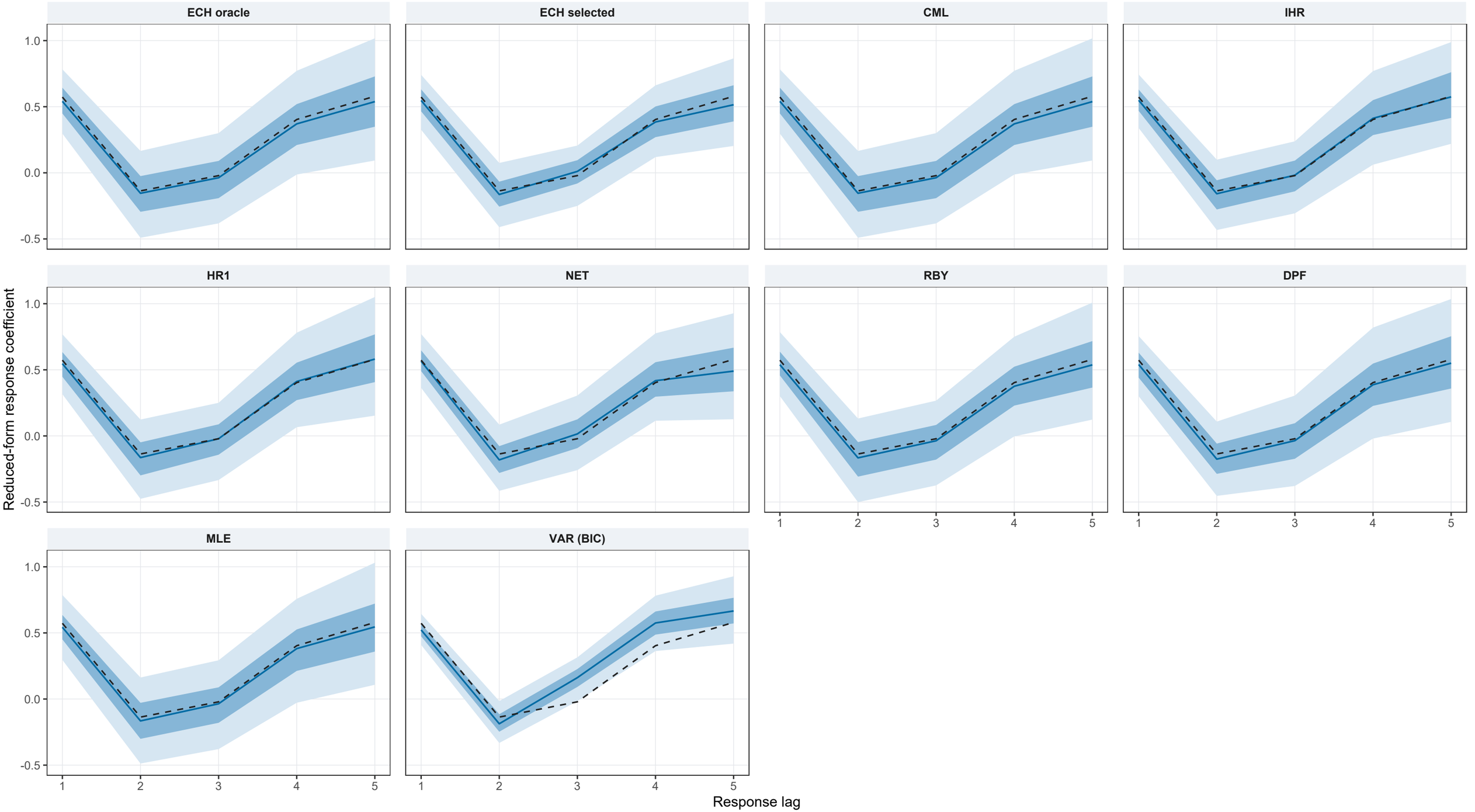
Qualified estimates | linear vertical axis | untrimmed RMSE



Errors are realized future observations minus forecasts. DGP uses true parameters and the same observed training history.
Qualified means stable AR, invertible MA and reported convergence where applicable; NET uses its IHR pilot's flag. Counts are in the diagnostic tables.

Design 4: response y1 to innovation 1

Sampling distribution of IRF estimates | n = 100



Blue line: median. Dark band: 25th-75th percentiles. Light band: 5th-95th percentiles. Dashed line: population response.
Bands describe variation across Monte Carlo replications; they are not confidence intervals. Extreme losses remain in the RMSE summaries.